Further consequences of the retuned translated-probe source

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Abstract

The source accompanying the conditional bound $\Lambda < 0.116398$ has two additional uses. A separately audited comparison shows that the zero-free hypothesis $\mathsf{H}_{34/43}$ suffices for $\Lambda < 0.09$. An exact adjacent-covariance argument supplies a weaker sufficient arithmetic cap, 3/16, on the same dyadic scale range as the earlier 1/6 cap. That all-scale cap is not proved here. A final result limits comparisons using only the older frozen source floors: even after their constraints and own-pair costs are relaxed, they cannot certify extinction by 0.10875. That limitation is not a lower bound for Λ. These statements are separate from the main 7/8 conditional theorem and are not its premises.

1 Notation and evidence

Use the heat-flow normalisation, source and first-contact proof of the main paper [1]. Write H_α for the assertion that every nontrivial zeta zero has real part at most α. Each scalar step satisfies $q' \le -2 - cq$, with

$$c = \frac{2\lambda}{h}\left(\ell - \frac{M}{h}\right) > 0. \tag{1}$$

The source is `evidence/SOURCE_SELECTED.json` inside the evidence archive. No stronger initial strip is inferred from $\mathsf{H}_{7/8}$. The prior corpus audit [3] is a research document; its earlier endpoints are historical and do not replace the main paper's bound.

2 A weaker arithmetic supplier for $\Lambda < 0.09$

The new source can also be consumed with a different initial strip. This is a separate implication, not a deduction from $\mathsf{H}_{7/8}$.

Theorem 1 (The 34/43 supplier)**.** *Assume* $\mathsf{H}_{34/43}$. *Then the same selected physical source gives*

$$\Lambda \le \frac{89979281294859}{10^{15}} = 0.089979281294859 < 0.09. \tag{2}$$

Proof. The time-zero squared height is at most $(25/43)^2 = 625/1849$. Choose $q_0 = 625/1849 + 10^{-6} = 625001849/1849000000$ and retain all the other comparison rules. Both separately audited histories contain 11,518 steps, all at fixed shift $r = 3/5$. The alternate upper bound is 0.089979281294860. The global first-contact argument of the main paper [1] proves the result. □

For comparison, the earlier corpus audit used the stronger strip $\mathsf{H}_{11/14}$ as a sufficient input and obtained 0.089701909084726 [3]. Since $34/43 > 11/14$, the new supplier allows zeros farther from the critical line. No assertion is made that 34/43 is the largest possible sufficient strip.

2.1 The all-scale adjacent-covariance premise

Fix $X = 2^j$ and $L = 8X$. Regard the actual finite Möbius block $\mu(1), \dots, \mu(L)$ cyclically, subtracting its exact block mean $\bar{\mu} = L^{-1} \sum_{n=1}^{L} \mu(n)$. For $h \geq 1$, define

$$Y_h(i) = \sum_{r=1}^{h} (\mu(i+r) - \bar{\mu}), \quad E_X(h) = \sum_{i=1}^{L} Y_h(i)^2, \quad C_X(h) = \sum_{i=1}^{L} Y_h(i) Y_h(i+h),$$

with indices read modulo L. Put $H_X = \left\lceil X^{17/30} \right\rceil$ and $h_0 = 2^{\lfloor j/100 \rfloor}$.

Theorem 2 (Relaxed covariance implication)**.** *Suppose there exist fixed finite $B \geq 0$ and j_0 such that, for every $j \geq j_0$, the actual Möbius block satisfies*

$$\boxed{C_X(h) \leq \frac{3}{16} E_X(h) + BLh} \tag{3}$$

for all $h = 2^k$ with $\lfloor j/100 \rfloor \leq k < \lceil \log_2 H_X \rceil$. Then $\Lambda < 0.09$. The all-scale premise (3) *is not established here.*

Proof. The exact concatenation identity $Y_{2h}(i) = Y_h(i) + Y_h(i+h)$ and cyclic translation invariance give

$$E_X(2h) = 2E_X(h) + 2C_X(h).$$

Set $F_X(h) = E_X(h) + (16/3)BLh$. Under (3),

$$F_X(2h) \leq \frac{19}{8} F_X(h). \tag{4}$$

The constants match because $2 + 2(16/3) = (19/8)(16/3)$. The integer inequality $19^{125} < 2^{531}$ proves $19/8 < 2^{156/125}$. Write $p = 156/125$. Centring decreases the cyclic raw energy, so $E_X(h_0) \leq Lh_0^2$. Consequently, for eligible dyadic $h \geq h_0$,

$$F_X(h) \leq (1 + 16B/3) L h_0^{2-p} h^p.$$

Assemble the binary expansion of H_X using translated dyadic blocks. Minkowski's inequality for the cyclic ℓ^2 norm preserves these translations. The sum of the normalised dyadic square-root bounds is at most $(1 - 2^{-p/2})^{-1}$. A residual block shorter than h_0 costs at most one further unit. Because $p > 1$, their total is less than five. Using $h_0 \leq X^{1/100}$, $H_X \leq 2X^{17/30}$, $p < 2$ and $L = 8X$ gives

$$E_X(H_X) \leq 800(1 + 16B/3) X^{10717/6250}. \tag{5}$$

The bound $E_X(r) \leq Lr^2$ used for the residual follows from centring and the raw sums, not from the false pointwise assertion $|\mu(n) - \bar{\mu}| \leq 1$.

For the noncyclic windows with starts $X \leq x < 4X$, put

$$B_X(x) = \sum_{x < n \leq x + H_X} \mu(n), \quad b_X = \frac{1}{3XH_X} \sum_{x=X}^{4X-1} B_X(x), \quad W(X) = \sum_{x=X}^{4X-1} |B_X(x) - H_X b_X|^2.$$

All these windows lie inside the retained cyclic block for sufficiently large X. Minimising their own fitted mean gives $W(X) \leq E_X(H_X)$.

For any bounded coefficients $|a_n| \leq 1$, the finite sliding-window identity, its complete triangular edge charge, and Cauchy–Schwarz imply

$$|A(Y) - A(X) - b_X(Y - X)| \leq H_X - 1 + \frac{\sqrt{3XW(X)}}{H_X} =: \mathcal{E}_X \tag{6}$$

for $X \le Y \le 4X$, where $A(Y) = \sum_{n \le Y} a_n$. This is the overlapping reconstruction in [2, 3]. In the present case (5) yields

$$\mathcal{E}_X = O(X^\alpha), \qquad \alpha = \frac{1 + 10717/6250}{2} - \frac{17}{30} = \frac{29651}{37500} < 1.$$

Compare (6) at scales X and $2X$ on their overlap:

$$|b_{2X} - b_X| \le \frac{2\mathcal{E}_X + \mathcal{E}_{2X}}{2X}.$$

The dyadic slope differences are summable. They converge to a constant c, with $b_X - c = O(X^{\alpha-1})$. Telescoping the affine reconstruction then gives $A(Y) = cY + O(Y^\alpha)$.

For Möbius, the exact divisor identity $\sum_{d \le N} \mu(d) \lfloor N/d \rfloor = 1$ implies $|\sum_{d \le N} \mu(d)/d| \le 2$. Partial summation of $A(Y) = cY + O(Y^\alpha)$ would give $c \log N + O(1)$ for that harmonic sum. Thus $c = 0$ and $M(Y) = O(Y^{29651/37500})$. The Mellin integral $s \int_1^\infty M(u) u^{-s-1} \mathrm{d}u$ is holomorphic to the right of this exponent and equals $1/\zeta(s)$ for $\operatorname{Re} s > 1$. The identity theorem excludes zeta zeros in the larger half-plane. Finally,

$$\frac{34}{43} - \frac{29651}{37500} = \frac{7}{1612500} > 0,$$

so $\mathsf{H}_{34/43}$ follows, and Theorem 1 completes the implication. □

The earlier 1/6 covariance supplier [3] used the same $h_0 = 2^{\lfloor j/100 \rfloor}$ and the same dyadic range. Replacing 1/6 by 3/16 enlarges the permitted coefficient by a factor 9/8, with no extra small-scale requirement. This is a genuine weakening of a sufficient premise, but not evidence that the actual Möbius function satisfies it. Directly proving $W(X) \ll X^{1106/645}$ up to fixed logarithmic factors on every sufficiently large dyadic scale is another sufficient supplier.

3 A restricted limit of the old floor-table method

This section concerns the *old* fixed height per time cell and its recorded lower floors. It does not concern the new source, and it does not bound the true constant from below.

Proposition 3 (Frozen-table comparison limit)**.** *Consider comparisons obtained by nonnegative linear combinations of translated kernels at the old height h_i in each time cell. Assume they use only the recorded lower floors $\ell_{i,m}$, retain a nonnegative own-pair subtraction, and use the rate form in* (1)*. No such comparison can certify extinction by time* $87/800 = 0.10875$*.*

Proof. For fixed y, h, b and bounded translations, the far-field asymptotics are $K_y(d, b) \sim 2y/d^2$ and $J_{h,r}(d, b) \sim 2h/d^2$. Thus a kernel inequality $2K_y \ge (y/h) \sum_m w_m J_{h,r_m}$ with $w_m \ge 0$ forces $\sum_m w_m \le 2$. Even if every kernel-domain constraint and every own-pair cost is discarded, the rate is bounded above by

$$U_i = \frac{4}{h_i} \max_m \ell_{i,m},$$

where the maximum includes all recorded continuum bands. The fastest relaxed comparison therefore solves $R' = -2 - U_i R$ on the cell, with the ideal seed $R(0) = 9/16$. Every nonnegative comparison in the stated class lies above this one until extinction, by scalar comparison.

The supplied rational checker propagates a lower enclosure for R. A degree-19 odd Taylor polynomial bounds e^{-x} below for the checked $0 \le x < 1$; downward rounding uses denominator 2^{144}. After 435 complete cells it verifies

$$R(87/800) > 37/10^6 > 0, \qquad R(9/100) > 46/1000.$$

Hence the relaxed comparison, and therefore every comparison in the specified class, is still positive at 0.10875. □

The restriction to numerical floor information matters. A new physical lower bound, another height, a justified signed construction with upper as well as lower source bounds, or additional joint information changes the problem. Proposition 3 is neither a theorem that $\Lambda > 0.10875$ nor an obstruction to proving RH by a different argument. It shows why repeated scalar step refinement of the old table is not a route to 0.09.

4 Verification boundary

The complete audit checks all eight retained final-source histories, the rational constants in the covariance implication and the downward-rounded relaxed comparison for the old-table limitation. It does not prove the all-scale covariance premise. The scripts are `code/check_covariance_bridge.py`, `code/check_table_limit.py` and the separate scalar auditor. The *Verification Guide* specifies their execution and input binding.

References

[1] D. J. Raymond, *Retuned translated probes and the conditional bound* $\Lambda < 0.116398$ *for the de Bruijn–Newman constant*, version 1.0, 10 October 2026. Main paper in this deposit.

[2] D. J. Raymond, *Overlapping centred Möbius windows* and *Adjacent covariance*, 6 October 2026. The exact source proofs are retained in `context/` inside `Proof_and_Replay.zip`. Their all-scale arithmetic premises remain unproved.

[3] D. J. Raymond, *From Quasi-RH to the next QED*, research audit, 8 October 2026, Sections 5–7. Supplied historical source in `context/QRH_Corpus_QED_Audit_2026-10-08.pdf` inside the evidence.

[4] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann* ξ *function, and a new upper bound for the de Bruijn–Newman constant*, Research in the Mathematical Sciences **6** (2019), article 31; arXiv:1904.12438v2.